Spatial–Spectral Joint Reconstruction With Interband Correlation for Hyperspectral Anomaly Detection

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***Abstract*— Hyperspectral image (HSI) anomaly detection (AD) is an important task in the remote sensing domain. In recent years, many scholars have been addicted to constructing deep network-based methods for hyperspectral AD and have developed numerous related methods. Many of them are designed based on autoencoder, which aims to reconstruct a stable background to identify anomalies. However, these autoencoder-based methods suffer from some problems, such as ignoring the interband correlation in HSI, that is, the HSI presents spectral similarity as well as redundancy between the contiguous bands, which would affect the reconstruction of the HSI. Moreover, the current anomaly detectors lack the use of spatial contextual information that exists in the pixel neighbor region when constructing the detector. To tackle these problems, this study presents a spatial–spectral joint reconstruction with the interband correlation-based anomaly detector (denoted as SSRICAD) for HSIs. We first divide the original HSI into several sub-HSIs by a band cross-grouping strategy to reduce the redundancy and impose the interband correlation constraint (IBCC) into the reconstruction process. Then, an outlier removal constraint (ORC) is added to alleviate anomaly contamination, which could help rebuild a more stable and pure background component. Finally, spatial information is extracted from the pixel neighbor region to contribute to the spatial–spectral joint reconstruction and further enhance the detection performance. Extensive experiments on three benchmark hyperspectral datasets indicate that the proposed SSRICAD can achieve superior performance in AD.**

***Index Terms*— Anomaly detection (AD), deep learning, hyperspectral image (HSI), interband correlation, spatial–spectral joint reconstruction.**

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I. Introduction

NOWADAYS, with the successful development of hyperspectral spectrometers, hyperspectral remote sensing technology has become one of the frontier technologies for Earth observation [1]. By means of airborne or spaceborne sensors, hyperspectral images (HSIs) can capture hundreds of narrow contiguous bands, which provide high spectral resolution and convey rich spectral information [2]–[6]. This characteristic provides the possibility to observe and analyze various ground cover materials and to distinguish subtle spectral differences between them [7], [8]. As a result, HSI has been used in various fields. Among them, hyperspectral anomaly detection (AD) attracts great attention for its real-world applications, such as precision agriculture, civilian reconnaissance, and surveillance, and hyperspectral AD becomes an important and fundamental research topic in the remote sensing domain [9], [10].

Hyperspectral AD, which aims to distinguish and locate the anomaly targets in HSI, is an unsupervised method [11], [12]. The anomaly targets usually refer to those unusual materials in the HSI whose spectra are remarkably different from their circumambient homogenous background. Generally, anomaly targets are those small objects with low probabilities of occurrence [13]. Compared with supervised target detection [14], [15], hyperspectral AD may be a more challenging task since it has no prior information about anomalies or backgrounds. Generally speaking, hyperspectral AD can be modeled as an unsupervised binary classification problem as it labels pixels of the image as anomaly targets or backgrounds [16]. Nevertheless, due to the small proportion of anomaly targets in HSI, the overall classification error may influence the accuracy of hyperspectral AD. Therefore, classification theory is not suitable for solving AD problems. Generally, traditional anomaly detectors are formed based on the Neyman–Pearson criterion [12].

Since there is no prior information about anomalies or backgrounds, the current hyperspectral AD algorithms usually extract features from backgrounds of HSI and distinguish anomalies by the difference between them [17]. In the past decades, a series of classical hyperspectral AD methods have been developed. One of the benchmark AD methods is the Reed-X detector (RX) [18], which is a typical Mahalanobis

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distance-based method that assumes the background that obeys a multivariate Gaussian distribution and has been widely used for hyperspectral AD. Based on the RX, Schaum [19] later presented a subspace RX (SSRX), which combines joint subspace detection. To estimate the background covariance matrix more accurately, several modified RX methods have been introduced, such as uniform target detector (UTD) [20], local RX (LRX), and other weight RX [21], [22]. However, the multivariate Gaussian distribution assumption is not satisfied with most hyperspectral datasets in practice, thus causing a poor detection performance. After that, the kernel-based anomaly detectors were proposed based on the kernel theory. Typical methods are the kernel RX [23], the kernel principal component analysis (PCA) [24], support vector data description [25], and the robust nonlinear anomaly detector (RNAD) [26]. They enhance the separability of the anomalies and the backgrounds by projecting the raw data into a higher dimensional feature space. The detection performance of these kernel-based algorithms may perform well when the datasets meet certain statistical hypotheses, which are not satisfied in most cases.

Different from the aforementioned methods, representation-based methods have become the mainstream of hyperspectral AD in the past decades because they do not require any statistical hypothesis in advance [27], [28]. They use the concept that backgrounds are able to be represented by pixels in the spatial neighbor regions approximately, while anomalies cannot [29]. Based on this concept, two representation-based methods, i.e., collaborative representation-based detector (CRD) [29] and sparse representation-based detector (SRD) [30], are presented. They calculate the reconstruction errors of each pixel to distinguish anomalies. Following the CRD and SRD, many representation-based hyperspectral anomaly detectors have been presented. Zhang *et al.* [31] proposed a low-rank and sparse matrix decomposition anomaly detector (LSMAD). It tries to decompose the input image into the anomaly, background, anomaly, and noise components by the GoDec [32] algorithm to identify anomalies. Su *et al.* [33] presented a collaborative representation anomaly detector with outlier removal (PCACRD), which obtains the background component by PCA and outputs by the CRD detector. Qu *et al.* [34] proposed a low-rank decomposition detector based on spectral unmixing, which utilizes abundance vectors and the dictionary for low-rank decomposition.

In recent years, many scholars have been addicted to constructing deep network-based methods for hyperspectral AD. Li *et al.* [35] applied the deep learning-based method for hyperspectral AD for the first time and proposed a transferred anomaly detector based on the convolutional neural network (CNNAD). CNNAD extracts spectral deep features by convolutional operations to identify anomalies. However, CNNAD requires a referenced HSI with ground truth to train the deep network, which is not entirely reasonable for an unsupervised hyperspectral AD task. Therefore, state-of-the-art deep learning-based hyperspectral anomaly detectors are constructed based on unsupervised autoencoder (AE) networks. They identify anomalies by reconstructing the original HSI (ori-HSI) or forming an effective representation of the backgrounds. Lu *et al.* [13] proposed a manifold constrained AE-based anomaly detector, which exploits the embedding manifold of autoencoders and considers the global–local-based structure that tries to reveal inherent compositions of the HSI. The detector uses the latent representation with locality properties learned from the AEs to detect anomaly targets. Xie *et al*. [36] presented a method of learning spectral features in an adversarial way for hyperspectral AD using a simple AE to extract spectral features of the HSI and further distinguishing the backgrounds and anomalies by suppressing noise. Lei *et al.* [37] proposed an AE-based discriminative reconstruction with spectral learning (SLDR) detector. In SDLR, spectral mapper angle (SAM) [38] scores between the rebuild pixels and the original pixels are calculated for the original detection result. Then, SLDR refines it by a weight function and outputs the final result based on the Mahalanobis detector. In [39], an end-to-end local invariant autoencoding density estimation model is presented for hyperspectral AD, while a robust graph autoencoder-based anomaly detector is provided in [40].

The current AE-based detectors have shown their capability for hyperspectral AD. Nevertheless, for the AD task, most existing deep learning-based methods still suffer from some problems as follows.

1) HSI contains hundreds of contiguous bands with a high spectral resolution, which can provide rich spectral information and even presents spectral similarity as well as redundancy between the contiguous bands. The inherent spectral similarity is the interband correlation of HSI. Existing AE-based methods, however, ignore the interband correlation for reconstruction when forming detectors. Under this circumstance, the adjacent bands contain redundancy, which may affect the reconstruction of the HSI, thus resulting in poor detection performance.
2) Existing deep learning-based anomaly detectors mainly focus on rebuilding a stable and pure background and use the Mahalanobis detector or the difference between the reconstructed image and the ori-HSI to distinguish anomalies. The major challenge of these methods is to reduce anomaly contaminations and obtain a stable and pure background during reconstruction. Nevertheless, they neglect this procedure and fail to alleviate anomalies when designing the network.
3) Apart from abundant spectral information, HSI presents spatial contextual information in the pixel neighbor region, especially in heterogeneous regions. Existing deep network-based detectors seldom take advantage of spatial contextual information in the designed detector, which achieves unsatisfactory performance.

To address the above problems, we present a new anomaly detector based on spatial–spectral joint reconstruction with interband correlation (SSRICAD) for HSIs. The following summarizes the major contributions of this study.

1) A spatial–spectral joint reconstruction framework, which integrates the interband correlation, outlier removal, and spatial information constraint (SIC) into a unified network, is presented for hyperspectral AD.

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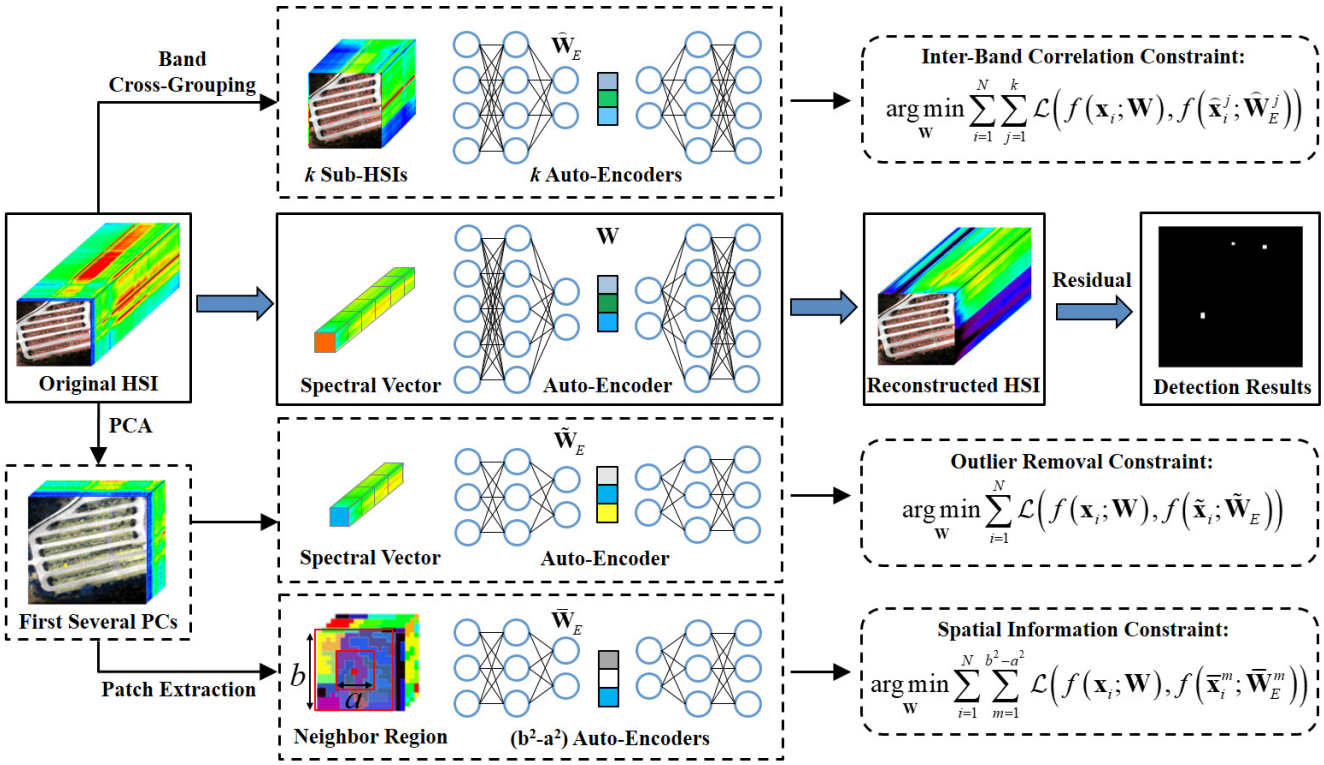


Fig. 1. Flowchart of the SSRICAD algorithm.

2) We investigate the inherent spectral similarity contained in adjacent bands of HSI. To reduce redundancy and avail of the interband correlation, latent features of sub-HSIs learned by well-designed AEs are applied to learn a latent representation of the ori-HSI more precisely, thus enhancing the discriminative ability between anomalies and backgrounds.
3) We present a simple but effective trick that uses the background component obtained by PCA to alleviate anomaly contaminations and impose an outlier removal constraint (ORC) into image reconstruction. This may be a promising way to deal with anomaly contaminations and help regain a more stable and pure background.
4) To utilize the spatial contextual information contained in the pixel neighbor region, we utilize the pixels in the dual concentric window to constrain the latent representation of the central pixel. In this way, the proposed SSRICAD can take advantage of spatial discriminative features, which could further contribute to the reconstruction of the HSI.

The rest of this study is organized as follows. Section II introduces the SSRICAD framework, including three main constraints. Section III provides the detection results and analyses conducted to verify the three constraint parts and the superiority of SSRICAD. Finally, conclusions are presented in Section IV.

II. SSRICAD Algorithm for HSI Anomaly Detection

In this section, the proposed SSRICAD for hyperspectral AD is detailedly presented. The flowchart of SSRICAD is shown in Fig. 1. As shown in Fig. 1, the input of SSRICAD is the single ori-HSI since there is no prior information about the anomaly and the background for hyperspectral AD. The goal of the SSRICAD is to extract knowledge from the backgrounds and get an efficient representation for the background as well as alleviate anomaly contaminations. Then, the SSRICAD distinguishes anomaly targets by the difference between the ori-HSI and the reconstructed image with a more stable and pure background. To this end, the SSRICAD enhances the reconstruction of the HSI from three aspects: 1) first, the interband correlation constraint (IBCC) is imposed to reduce the redundancy and avail of inherent spectral similarity, enhancing the discriminative ability between anomalies and backgrounds; 2) to alleviate anomaly contaminations, an ORC is added in the SSRICAD, in which main background components are used to help regain a more stable and pure background; and 3) an SIC that constrains the latent representation of the central pixel is imposed to utilize the spatial information that exists in the pixel neighbor region.

A. BaseNet of HSI Reconstruction for SSRICAD

Since there is no prior information about the anomaly and the backgrounds, the current deep learning-based hyperspectral anomaly detectors are constructed based on unsupervised networks, which typically choose the autoencoder (AE) network. The proposed SSRICAD is no exception. The AE network is a typical unsupervised network that aims to learn a latent representation of the input by making sure that input configurations can have a low reconstruction error [41], which can fit the HSI reconstruction of the SSRICAD well.

Consider the ori-HSI by $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^{N} \text{in} \mathbb{R}^B$, where N denotes the number of the total pixels (training samples) and B is the

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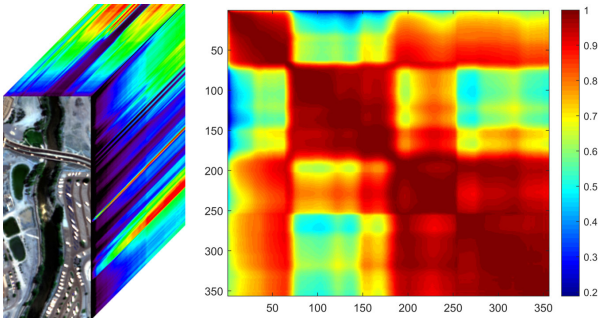


Fig. 2. Spectral similarity of the Reno Urban data. The left is the 3-D cube of the Reno Urban image, and the right is a correlation coefficient matrix between each single band image.

number of the spectral bands. The goal of the baseNet of the SSRICAD is to minimize

$$\arg\min_{\mathbf{W}} \sum_{i=1}^{N} \mathcal{L}(\mathbf{x}_i, f(\mathbf{x}_i; \mathbf{W})) \tag{1}$$

where $f(\mathbf{x}_i; \mathbf{W})$ is a function of the baseNet parameterized by $\mathbf{W}$ and $\mathcal{L}(\cdot)$ is a loss function, which chooses the mean square loss for the regression task of the SSRICAD in this article. In SSRICAD, each pixel is regarded as a spectral vector and is fed into the well-designed network, where the number of nodes in the first hidden layer is set as, B, 128 and 64 for the second and third hidden layers, respectively, and the dimension of the latent feature learned by the encoder is set as 32. By feeding the spectral vectors of the entire HSI into the network, the baseNet optimizes the network using the mini-batch stochastic gradient descent algorithm and finally outputs the reconstructed HSI. The following formulates the loss function of the baseNet:

$$\mathcal{L}(\mathbf{x}_i, f(\mathbf{x}_i; \mathbf{W})) = \|\mathbf{x}_i - f(\mathbf{x}_i; \mathbf{W})\|_2^2. \tag{2}$$

B. Interband Correlation Constraint for SSRICAD

HSI contains hundreds of contiguous bands and presents a high spectral resolution. Due to this property, HSI even presents a great inherent spectral similarity as well as redundancy between the contiguous bands [42]. To better illustrate this inherent spectral similarity or redundancy of HSI, the Reno Urban hyperspectral dataset collected by the ProSpecTIR DUAL sensor is taken as an example. The following experiment part will provide a detailed introduction to the Reno Urban dataset. We calculate the similarity between each single band image of the HSI data via correlation coefficients. The inherent spectral similarity of the Reno Urban data is shown in Fig. 2, where the left is the 3-D cube of the data and the right is a correlation coefficient matrix between each single band image.

In Fig. 2, the axes represent the band order of the Reno Urban dataset. From Fig. 2, we can see that the correlation coefficient values near the diagonal are larger than the others and close to 1. Moreover, there is a red "block effect" near the diagonal of the correlation coefficient matrix, which indicates that several bands near others have a large correlation with each other, that is, HSIs with hundreds of bands present a great inherent spectral similarity (also called interband correlation) as well as high redundancy between the adjacent bands.

Under this circumstance, the adjacent bands of HSI contain redundancy, which may affect the reconstruction of the HSI, thus resulting in poor detection performance. To take advantage of the inherent spectral similarity and mitigate the negative impact of the redundancy, an IBCC is imposed for SSRICAD. First, the ori-HSI is divided into k sub-HSIs via a band cross-grouping strategy, denoted as $\{\widehat{\mathbf{X}}^j\}_{j=1}^k$ in $\widehat{B}$, where $\widehat{B}$ represents the number of bands in sub-HSI. Specifically, multiple adjacent bands of the holistic HSI are cross-grouped into several different groups according to the band order of the ori-HIS and different sub-HSIs are formed. For convenience, $\widehat{B}$ is calculated by $\widehat{B} = \text{floor}(B/k)$, where floor$(\cdot)$ denotes rounding down operation, that is to say, the number of bands in each sub-HSIs equals each other and the remanent bands are discarded. Since an HSI contains hundreds of bands, it is reasonable to discard few bands, which have little impact on the detection task. Taking the Reno Urban image data for an example, there are 356 bands in total. Suppose that we group the HSI into three sub-HSIs ($k = 3$), the number of bands in each sub-HSIs is 118 and the two remaining bands are discarded. The single bands labeled as $1, 4, 7, \ldots, 349, 352$ are formed as the first sub-HSI, and the single bands labeled as 2, 5, 8, ..., 350, 353, and 3, 6, 9, ..., 351, 354 are formed as the second and the third sub-HSI, respectively. In this way, the ori-HSI can be grouped into several highly related sub-HSIs because the corresponding bands in the same order of each sub-HSI have large correlation coefficients, such as the bands labeled as 1, 2, and 3. Besides, compared with the ori-HSI, these sub-HSIs contain less redundancy.

Similar to the reconstruction of the ori-HSI, k sub-HSIs are reconstructed using well-designed AEs, which can be denoted as

$$\arg\min_{\widehat{\mathbf{W}}^j} \sum_{i=1}^{N} \mathcal{L}\left(\widehat{\mathbf{x}}_i^j, f\left(\widehat{\mathbf{x}}_i^j; \widehat{\mathbf{W}}^j\right)\right) \tag{3}$$

where $\widehat{\mathbf{W}}^j$ are the network parameters for the jth sub-HSI and $j \in (1, k)$. In SSRICAD, the networks of sub-HSIs do not share the same parameters since each sub-HSI contains different information. In these AEs for the reconstruction of sub-HSIs, $\widehat{B}$ dimension spectral vectors are fed into the network to extract spectral features of the sub-HSIs, where the number of nodes in the first hidden layer is set as $\widehat{B}$ and others equal to that in the baseNet. The latent feature learned by the jth encoder is formulated as $f(\widehat{\mathbf{x}}^j; \widehat{\mathbf{W}}_E^j)$, where $\widehat{\mathbf{W}}_E^j$ refers to the parameter of the encoder in AE for the jth sub-HSI.

Now, how to combine these highly related sub-HSIs with the ori-HSI becomes a key issue. Our goal is to take advantage of the interband correlation and mitigate the negative impact of the redundancy for improving the reconstruction of HSI. An intuitive way is to make the features extracted from the ori-HSI similar to those extracted from the sub-HSIs. Inspired

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by this, we formulate the IBCC as

$$\arg\min_{\mathbf{W}} \sum_{i=1}^{N} \sum_{j=1}^{k} \mathcal{L}\left(f(\mathbf{x}_i; \mathbf{W}), f\left(\widehat{\mathbf{x}}_i^j; \widehat{\mathbf{W}}_E^j\right)\right). \tag{4}$$

It is obvious that (4) associates the sub-HSIs with the ori-HSI. In this way, not only the impact of the redundancy can be alleviated during the reconstruction of ori-HSI since the sub-HSIs contain less redundancy, but also the SSRICAD can take advantage of the inherent spectral similarity as these sub-HSIs are highly related.

C. Outlier Removal Constraint for SSRICAD

SSRICAD aims to reduce possible anomaly contaminations when reconstructing the ori-HSI and get a stable and pure background. However, how to reduce anomalies during reconstruction is a typical challenge for AD. Therefore, designing an applicable model that can purify the background to our problem is nontrivial.

Recently, it has been proven that detection tasks can benefit from principal components (PCs) of HSI [33], [43]. Su *et al.* [33] introduced a collaborative representation detector with outlier removal, where background PCs are utilized to collaboratively represent the central pixel and contribute to identifying anomalies. Yang *et al.* [43] proposed a data-augmented matched subspace detector, in which principal discriminative background information is extracted from target-background mixed subspace. Motivated from these works, we apply PCA to the ori-HSI and get background PCs to alleviate possible anomaly contaminations and then present an ORC for SSRICAD and formulate the constraint as

$$\arg\min_{\mathbf{W}} \sum_{i=1}^{N} \mathcal{L}\left(f(\mathbf{x}_i; \mathbf{W}), f\left(\tilde{\mathbf{x}}_i; \tilde{\mathbf{W}}_E\right)\right) \tag{5}$$

where $\tilde{\mathbf{X}} = \{\tilde{\mathbf{x}}_i\}_{i=1}^{N} \text{in} \mathbb{R}^p$ is the first several PCs derived from the ori-HSI via PCA, $\tilde{\mathbf{W}}_E$ denotes the network parameters of the encoder for the ORC, and p is the number of bands of selected PCs. In this study, we set $p = 5$, where the PCs still have useful discriminative information and there are few anomaly contaminations at the same time.

From (5), we can know that the objective of the ORC is to make the features extracted from the ori-HSI approach that of the background PCs, thus helping alleviate possible anomaly contaminations and leading to a purer background, which is beneficial for AD. Moreover, the process of PCA is an operation to remove redundancy, which can also mitigate redundancy and finally improve the reconstruction of the ori-HSI.

D. Spatial Information Constraint for SSRICAD

Due to the low spatial resolution of HSIs, most of the existing deep network-based methods seldom take advantage of spatial contextual information when designing the detector. However, HSIs present some spatial contextual information in the pixel neighbor region, especially in heterogeneous regions, which is favorable for detection. To take advantage of spatial contextual information and contribute to the spatial–spectral joint reconstruction, we develop an SIC for the SSRICAD.

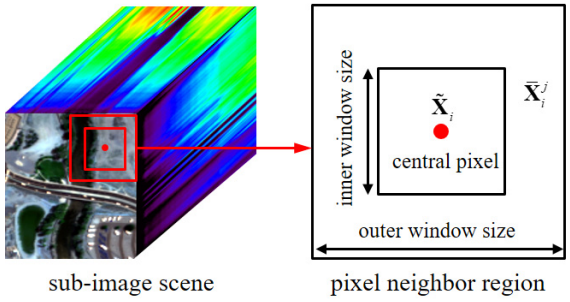


Fig. 3. Illustration of the dual window for pixel neighbor region with the Reno Urban data.

First, the image is divided into numerous pixel neighbor regions by the dual window (see Fig. 3) [44], [45] that contains a large outer window and a small inner window. Each pixel in the image owns a pixel neighbor region. In order to reduce the computational costs, the same PCs $\tilde{\mathbf{X}} = \{\tilde{\mathbf{x}}_i\}_{i=1}^{N}$ as the ORC portion, derived from the ori-HSI via PCA, are used in this section. An illustration of the dual window for pixel neighbor region with the Reno Urban data is shown in Fig. 3. Specifically, let $\tilde{\mathbf{x}}_i$ be the central pixel that corresponds to $\mathbf{x}_i$ in the ori-HSI, and pixels falling into the outer region of the dual window ($\bar{\mathbf{x}}_i^{ji}$ in Fig. 3) can be denoted as $\{\bar{\mathbf{x}}_i^m\}_{m=1}^{b^2-a^2}$, where a is the inner window size and b refers to the size of the outer window. In SSRICAD, we use these pixels to form SIC and formulate it as

$$\arg\min_{\mathbf{W}} \sum_{i=1}^{N} \sum_{m=1}^{b^2-a^2} \mathcal{L}\left(f(\mathbf{x}_i; \mathbf{W}), f\left(\bar{\mathbf{x}}_i^m; \bar{\mathbf{W}}_E^m\right)\right) \tag{6}$$

where $\bar{\mathbf{W}}_E^m$ denotes the network parameters of the mth encoder for the SIC.

It can be seen in (6) that the feature of the pixel $\mathbf{x}_i$ corresponding to the central pixel $\tilde{\mathbf{x}}_i$ tends to be closer to those pixels in the outer region. For each local region, pixels in the outer window are more likely to model the local background, whereas the inner window is for removing anomaly pixels. By applying (6), the reconstructed HSI will be more likely a pure background since each reconstructed central pixel is similar to its outer region pixels. In this way, the SSRICAD takes advantage of the spatial contextual information and further alleviates anomalies, which can enhance the final detection performance.

E. Framework of the Proposed SSRICAD

According to the aforementioned descriptions, IBCC, ORC, and SIC are three important portions of the ori-HSI reconstruction. The three constraint portions can be further detailed as follows:

$$\begin{aligned}&\mathcal{L}^{\text{IBCC}}\left(f(\mathbf{x}_i; \mathbf{W}), f\left(\widehat{\mathbf{x}}_i^j; \widehat{\mathbf{W}}_E^j\right)\right)\\ &= \left\| f(\mathbf{x}_i; \mathbf{W}) - f\left(\widehat{\mathbf{x}}_i^j; \widehat{\mathbf{W}}_E^j\right)\right\|_2^2\end{aligned} \tag{7}$$

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$$\mathcal{L}^{\mathrm{ORC}}\left(f(\mathbf{x}_i;\mathbf{W}), f\left(\tilde{\mathbf{x}}_i;\tilde{\mathbf{W}}_E\right)\right) = \left\|f(\mathbf{x}_i;\mathbf{W}) - f\left(\tilde{\mathbf{x}}_i;\tilde{\mathbf{W}}_E\right)\right\|_2^2 \quad (8)$$

$$\mathcal{L}^{\mathrm{SIC}}\left(f(\mathbf{x}_i;\mathbf{W}), f\left(\bar{\mathbf{x}}_i^m;\bar{\mathbf{W}}_E^m\right)\right) = \left\|f(\mathbf{x}_i;\mathbf{W}) - f\left(\bar{\mathbf{x}}_i^m;\bar{\mathbf{W}}_E^m\right)\right\|_2^2. \quad (9)$$

To enhance the reconstruction of the ori-HSI and get a stable and pure background, these three constraint portions are added to the baseNet. The complete loss function of the SSRICAD is formulated as

$$\begin{aligned}\mathcal{L} &= \mathcal{L}^{\mathrm{base}} + \mathcal{L}^{\mathrm{IBCC}} + \mathcal{L}^{\mathrm{ORC}} + \mathcal{L}^{\mathrm{SIC}} \\ &= \frac{1}{N}\sum_{i=1}^{N}\|\mathbf{x}_i - f(\mathbf{x}_i;\mathbf{W})\|_2^2 \\ &\quad + \frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{k}\left[\left\|\widehat{\mathbf{x}}_i^j - f\left(\widehat{\mathbf{x}}_i^j;\widehat{\mathbf{W}}^j\right)\right\|_2^2 + \left\|f(\mathbf{x}_i;\mathbf{W}) - f\left(\widehat{\mathbf{x}}_i^j;\widehat{\mathbf{W}}_E^j\right)\right\|_2^2\right] \\ &\quad + \frac{1}{N}\sum_{i=1}^{N}\left[\left\|\tilde{\mathbf{x}}_i - f\left(\tilde{\mathbf{x}}_i;\tilde{\mathbf{W}}\right)\right\|_2^2 + \left\|f(\mathbf{x}_i;\mathbf{W}) - f\left(\tilde{\mathbf{x}}_i;\tilde{\mathbf{W}}_E\right)\right\|_2^2\right] \\ &\quad + \frac{1}{N}\sum_{i=1}^{N}\sum_{m=1}^{b^2-a^2}\left[\left\|\bar{\mathbf{x}}_i^m - f\left(\bar{\mathbf{x}}_i^m;\bar{\mathbf{W}}^m\right)\right\|_2^2 + \left\|f(\mathbf{x}_i;\mathbf{W}) - f\left(\bar{\mathbf{x}}_i^m;\bar{\mathbf{W}}_E^m\right)\right\|_2^2\right] \end{aligned} \quad (10)$$

where $\widehat{\mathbf{W}}^j$ is the network parameters of the jth sub-HSI reconstruction for IBCC, $\tilde{\mathbf{W}}$ denotes the parameters for ORC, and $\bar{\mathbf{W}}^m$ refers to the parameters of the mth neighbor region for SIC. We take advantage of the mini-batch stochastic gradient descent algorithm to optimize the parameters at the same time.

After obtaining the reconstructed HSI $\hat{\mathbf{X}} = \{\hat{\mathbf{x}}_i\}_{i=1}^N$, the SSRICAD detects anomaly pixels utilizing the difference between the ori-HSI and the reconstructed HSI by calculating the residual

$$D_{\mathrm{SSRICAD}}(\mathbf{x}) = \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|_2. \quad (11)$$

The pixels with a large residual value are classified as anomaly pixels; otherwise, they are background pixels. The implementation details of SSRICAD are shown in Algorithm 1.

Algorithm 1 SSRICAD for HSI Anomaly Detection

Input:

1) A HSI with samples $\{\mathbf{x}_i\}_{i=1}^N$
2) Parameters: the number of subsets k, the number of PCs p, the size of dual window b, a

Steps:

1) Apply the band cross-grouping strategy to the HSI and obtain k subsets $\left\{\widehat{\mathbf{X}}^j\right\}_{j=1}^k$.
2) Apply PCA to the HSI and reserve the first p PCs $\{\tilde{\mathbf{x}}_i\}_{i=1}^N$.
3) Extract pixels falling into the outer region of dual window with the size of $(b\text{-}a)\times(b\text{-}a)\times p$ in the dimension-reduced HSI.
4) Impose the IBCC, ORC, and SIC to the baseNet via (4), (5), and (6), respectively.
5) Input the training samples to the network and optimize (10) by the mini-batch stochastic gradient descent algorithm.
6) Obtain the reconstructed HSI $\{\hat{\mathbf{x}}_i\}_{i=1}^N$.
7) For each pixel, calculate the residual via (11).

Output:

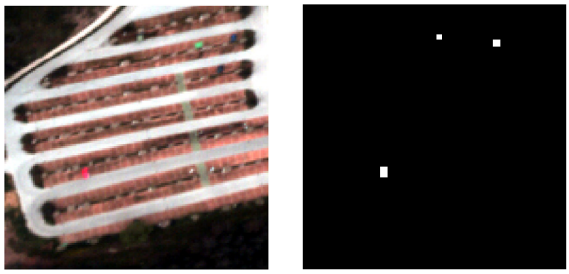
A two-dimensional image with the detection results.

III. Numerical Experiments and Analyses

In this section, to evaluate the effectiveness of the proposed SSRICAD, numerous experiments are conducted on three public hyperspectral datasets. We compare the SSRICAD with several advanced AD methods. Besides, we perform numerical experiments to analyze model parameters, ablation studies, and computational costs for SSRICAD.

A. HSI Dataset Description

The first HSI is an avionic pushbroom hyperspectral sensor (called the SIM.GA sensor) collected image, which is obtained during an HSI collection campaign named the "Viareggio 2013 trial" [46]. It can be downloaded from http://rsipg.dii.unipi.it/. Among the collected dataset in this campaign, the image in D1F12H1_Cropped is selected. The image scene is shown in Fig. 4(a), covering a parking lot and several vehicles. This dataset includes 511 spectral bands. The spectral resolution of this HSI is 2.3 nm with a wavelength range of 400–1000 nm and a spatial resolution of 0.6 m/pixel. The dataset has a size of 150 × 140 pixels, where three vehicles are regarded as anomalous materials. Fig. 4(b) shows the ground truth of this dataset, where three vehicles with 49 anomaly pixels need to be detected. The number of anomaly pixels accounts for 0.233% of the total pixels in this dataset.

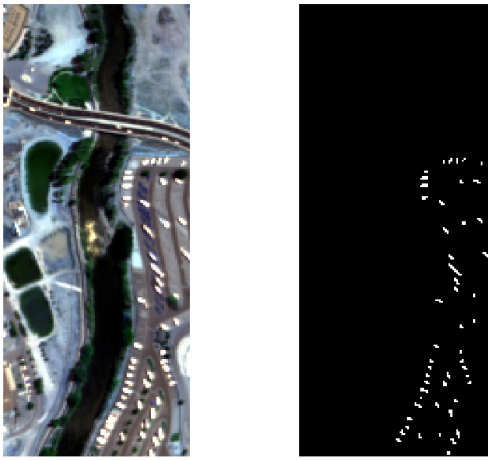


(a) (b)

Fig. 4. Viareggio dataset: (a) image scene and (b) ground truth.

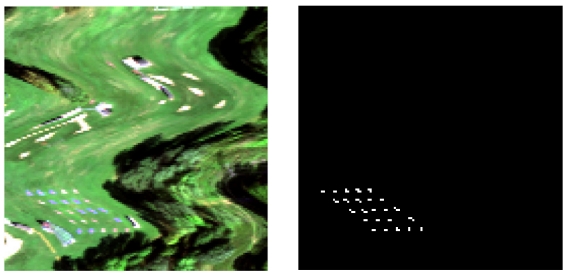
The second public dataset is a ProSpecTIR DUAL sensor collected HSI. The image scene of the dataset is shown in Fig. 5(a), covering an urban area in Reno, NV, USA (later called Reno Urban dataset) [47]. This image contains 356 spectral bands and 250 × 100 pixels. The ground truth of this dataset is shown in Fig. 5(b), where some vehicle objects containing 214 anomaly pixels are for detection. This HSI is downloaded from http://www.spectir.com/free-data-samples/.

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(a) (b)

Fig. 5. Reno Urban dataset: (a) image scene and (b) ground truth.



(a) (b)

Fig. 6. Avon dataset: (a) image scene and (b) ground truth.

The number of anomaly pixels accounts for 0.856% of the total pixels in this dataset.

The third benchmark dataset was obtained in the "SpecTIR Hyperspectral Airborne Experiment 2012" campaign by a pushbroom sensor [48], which can be downloaded from https://www.rit.edu/cos/share2012/downloads.php. The image scene of this dataset is a part of the park near a small town called Avon in the south of Rochester, USA, as shown in Fig. 6(a). In this dataset, there are 360 spectral bands with a wavelength range of 400–2450 nm and a spectral resolution of 5 nm. The spatial resolution of the dataset is 5 m/pixel. In this HSI, a scene covering 130×130 pixels is utilized as the experimental area. The ground truth of this dataset is presented in Fig. 6(b), where several panels containing 60 anomaly pixels are to be detected. The number of anomaly pixels accounts for 0.355% of the total pixels in this HSI.

B. Baselines

The following hyperspectral AD methods are taken as baselines for comparison.

1) RX [18] is a typical Mahalanobis distance-based method that distinguishes anomalies by calculating the Mahalanobis distance.
2) RNAD [26] is a kernel-based method that detects anomalies based on the kernel techniques.
3) CRD [29] recognizes anomalies as those pixels that cannot be represented by pixels in their neighbor region.
4) LSMAD [31] decomposes the input image into the anomaly, background, and noise components, focusing on the sparse component to detect anomaly pixels.
5) PCACRD [33] applies collaborative representation theory to the dimension-reduced image derived from HSI via PCA to distinguish anomalies.
6) CNNAD [35] trains a binary classifier with convolutional neural networks to classify anomalies and backgrounds according to a referenced HSI with ground truth.
7) SLDR [37] trains a deep autoencoder network using spectral information to reconstruct HSI and outputs detection results based on the Mahalanobis distance.

C. Experimental Settings

1) Evaluation Index: To evaluate the effectiveness of the SSRICAD, three widely used and valid criteria are adopted in this study, i.e., the receiver operating characteristic curve (ROC) [49], the area under the curve (AUC), and the separability map between anomaly and background [1], [50].

The ROC curve provides a threshold-free detection performance comparison by plotting the relation between the false alarm rate (FAR) and the probability of detection (PD). Specifically, PD can be formulated as N_d/N_t, where N_d is the number of detected anomalies and N_t denotes the count of real anomalies, while FAR is formulated as N_m/N, where N_m refers to the count of misjudged anomalies. Generally, the detector that can obtain a lower FAR under the same PD indicates that it has a better detection performance.

However, there is usually more than one curve that needs to be compared. When the ROC curves alternate, it could be difficult to judge which detector performs better. Under this circumstance, the AUC values compassed by the FAR axis and the curve are calculated for further evaluation. Apparently, a good detector can reach a large AUC value.

To further evaluate the separability ability of the SSRICAD method, the separability map that can visually present the separation between anomaly and background is adopted in this article. To form the separability map, we first normalize the detection results of the detector from 0 to 1 to calculate the normalized detection statistic range. Then, according to the ground truth, the detection results are divided into two categories corresponding to anomaly class and background class. Each class with the normalized detection statistic range forms a box, containing the main portions of the detection statistic results from 25% to 75%. Generally, the detector that can highlight anomalies and suppress backgrounds at the same time indicates good separability ability.

2) Parameter Settings: The SSRICAD has several critical parameters. The first parameter is the number of the subsets k, and we will provide a detailed parameter analysis of k in the following. For the number of PCs p, we set $p = 5$ in this article, where it still has useful discriminative information while removing anomaly contaminations. For the size of dual window, we empirically set $a = 5$ and $b = 13$ for the Viareggio

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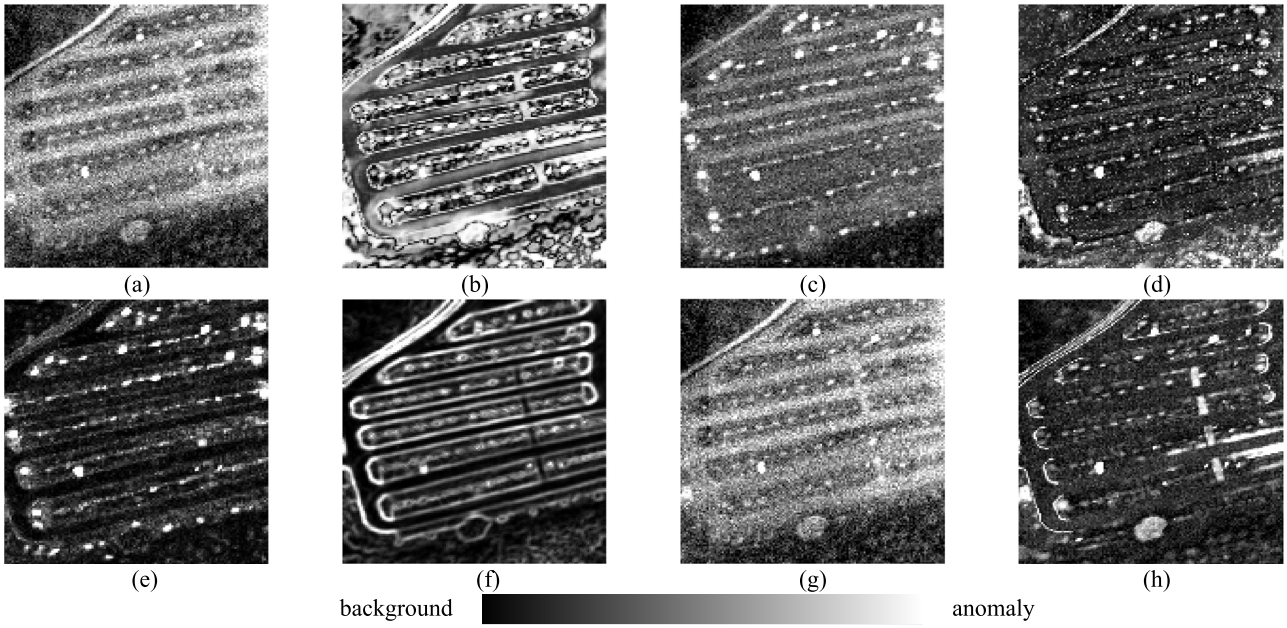


Fig. 7. AD result comparison of different methods for the Viareggio dataset: (a) RX, (b) RNAD, (c) CRD, (d) LSMAD, (e) PCACRD, (f) CNNAD, (g) SLDR, and (h) SSRICAD.

dataset and $a = 3$ and $b = 9$ for the Reno Urban and the Avon dataset, which can obtain robust performance. To train and fit the well-designed network, the learning rate is set as $1e^{-3}$ and the batch size is set as 128 in the proposed SSRICAD.

D. Analysis of Detection Performance

In this section, we present the comparison performance with state-of-the-art methods to evaluate the effectiveness of the proposed SSRICAD. First, we draw the detection result maps of the different methods for three datasets, as shown in Figs. 7–9. For the Viareggio dataset, RNAD and CNNAD hardly work in this dataset as they fail to suppress the background and also misjudge anomaly pixels. The methods, i.e., RX, CRD, LSMAD, PCACRD, and SLDR, can highlight partial anomalies correctly. For the Reno Urban dataset, RNAD still fails to restrain the background effectively, regarding most of the background pixels as anomaly pixels, while the other methods, i.e., RX, CRD, PCACRD, SLDR, and SSRICAD, can suppress them. Although SSRICAD performs a worse background suppression than RX, CRD, and PCACRD, it provides the most sufficient recognition of the anomalies among these detectors since it can reduce redundancy and avail of the interband correlation. For the Avon HSI, all the methods can identify anomalies precisely but CNNAD. Nevertheless, the bad effect of the suppression of the background materials in the middle of the image scene has an impact on the detection performance of these methods. However, it is difficult to verdict which detector performs better just from visual maps of detection results. Therefore, providing a quantitative detection performance comparison is needed.

The ROC curves' comparison for the three datasets is presented in Fig. 10, where the red curve draws the ROC curve of the proposed SSRICAD. For the Viareggio dataset, RNAD and CNNAD achieve bad performance as their ROC curves lie in the right-bottom corner. The SSRICAD outperforms others as it generates low FAR and high PD. For the Reno Urban dataset, the proposed SSRICAD has a prominent advantage as its curve always lies above other methods, while the ROC curves of the others alternate with each other. For the Avon dataset, CNNAD performs higher FAR under the same PD than other compared methods. ROC curve of the SSRICAD lies above all the others especially when the FAR is less than 0.01, indicating a superior detection performance. SSRICAD makes full use of the spatial contextual information as well as spectral information, and it can contribute to the reconstruction of the HSI, which can further enhance the detection performance.

To further compare the detection performance, we provide an AUC values comparison for the three datasets in Fig. 11. The performance of RX is not bad in the three datasets, while the kernel-based method RNAD shows a small AUC value in the Viareggio dataset. For the three representation-based detectors, i.e., CRD, LSMAD, and PCACRD, PCACRD works better in the Viareggio dataset, LSAMD outperforms in the Reno Urban image, while CRD achieves a better detection result in Avon HSI. For the deep learning-based methods, i.e., CNNAD and SLDR, they obtain fairly good performance in the Reno Urban dataset. However, CNNAD hardly works in the Viareggio and Avon datasets. Overall, the proposed SSRICAD has the largest AUC value, denoting the optimal detection performance among these compared methods for all three datasets.

Furthermore, to evaluate the separability ability of the proposed SSRICAD, we present a separability map comparison for the three datasets in Fig. 12. In separability maps, the green box illustrates the range of anomalies, while the orange box presents that of backgrounds. From Fig. 12, the methods, i.e., CRD, LSMAD, SLDR, and SSRICAD, can restrain the main

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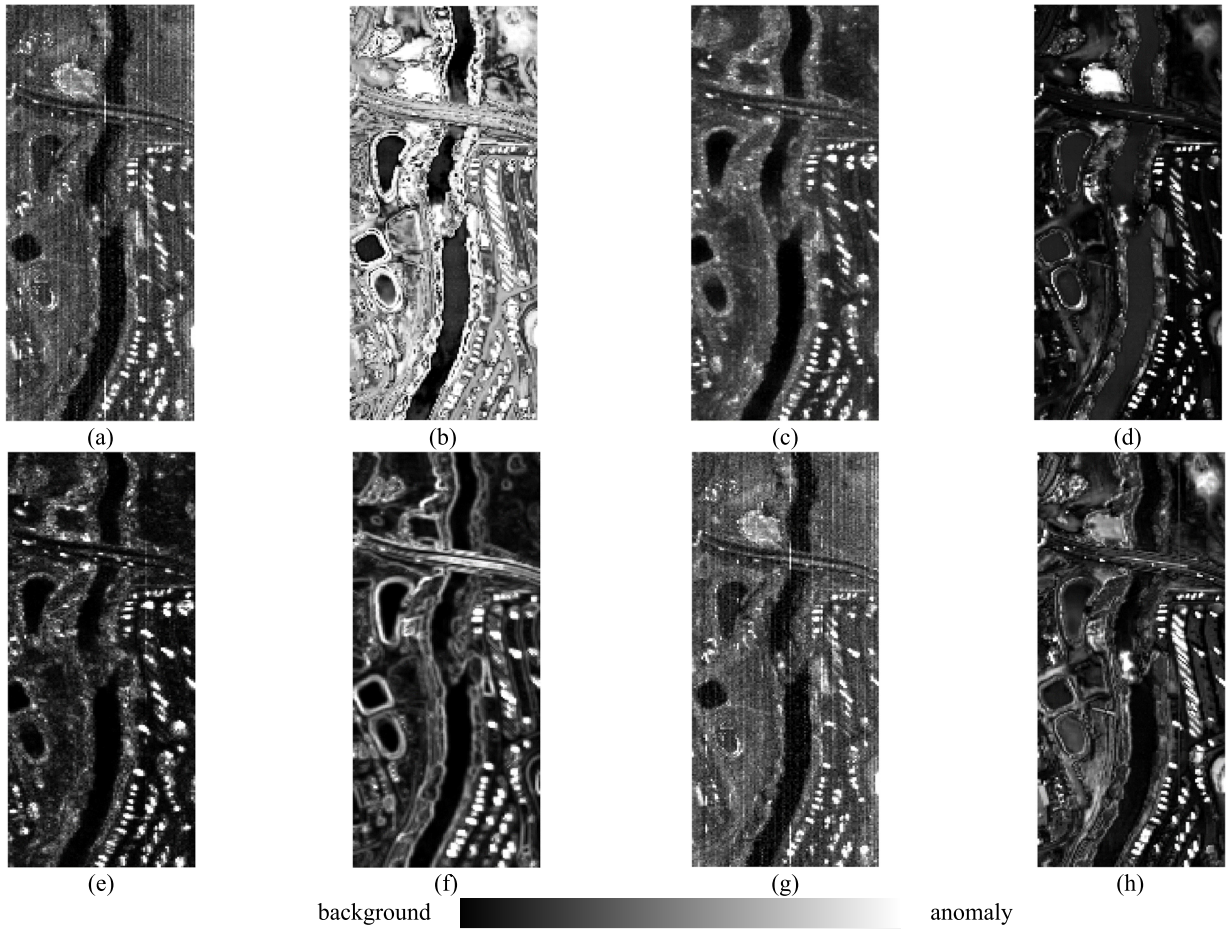


Fig. 8. AD result comparison of different methods for the Reno Urban dataset: (a) RX, (b) RNAD, (c) CRD, (d) LSMAD, (e) PCACRD, (f) CNNAD, (g) SLDR, and (h) SSRICAD.

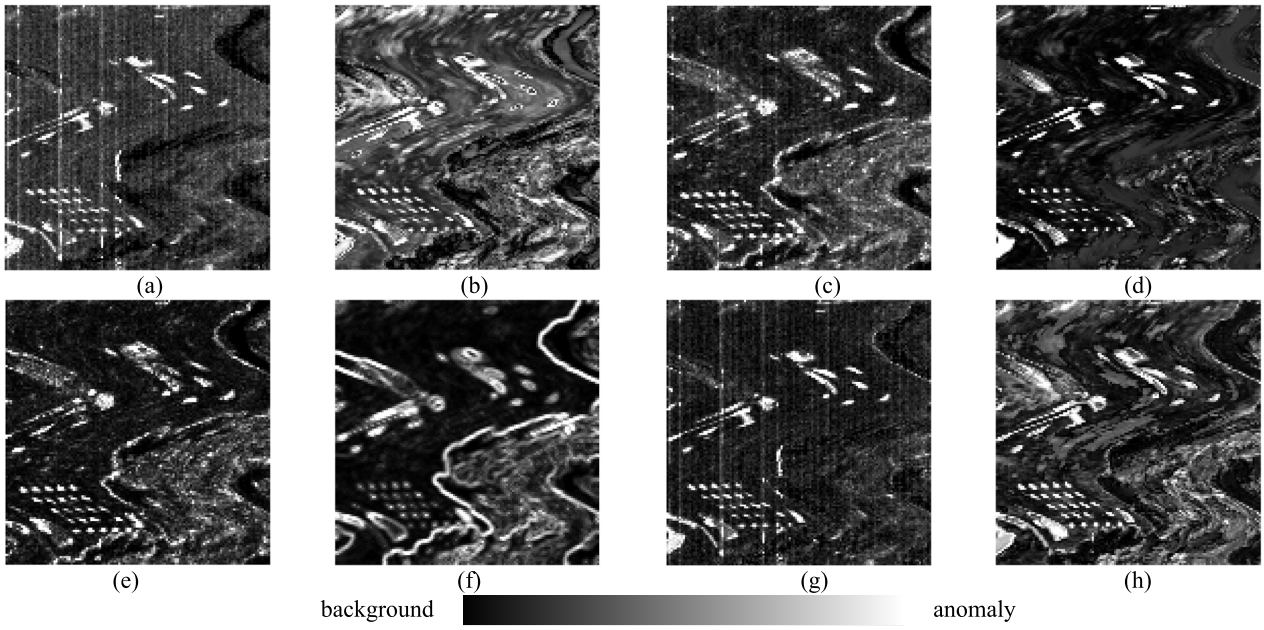


Fig. 9. AD result comparison of different methods for the Avon dataset: (a) RX, (b) RNAD, (c) CRD, (d) LSMAD, (e) PCACRD, (f) CNNAD, (g) SLDR, and (h) SSRICAD.

portion of backgrounds into a small scale of values for the Viareggio dataset. Among them, the proposed SSRICAD performs better in highlighting anomalies, which indicates a better separation between anomaly and background. For the Reno Urban dataset, the detectors, i.e., RX, CRD, PCACRD, and SLDR, work well in background suppression. Although the

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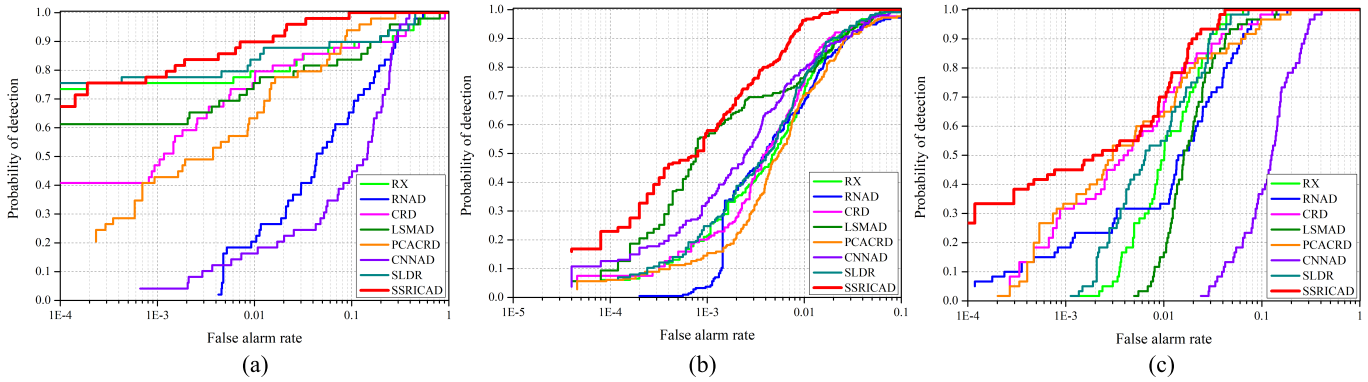


Fig. 10. Comparison of ROC performance for three datasets: (a) Viareggio, (b) Reno Urban, and (c) Avon.

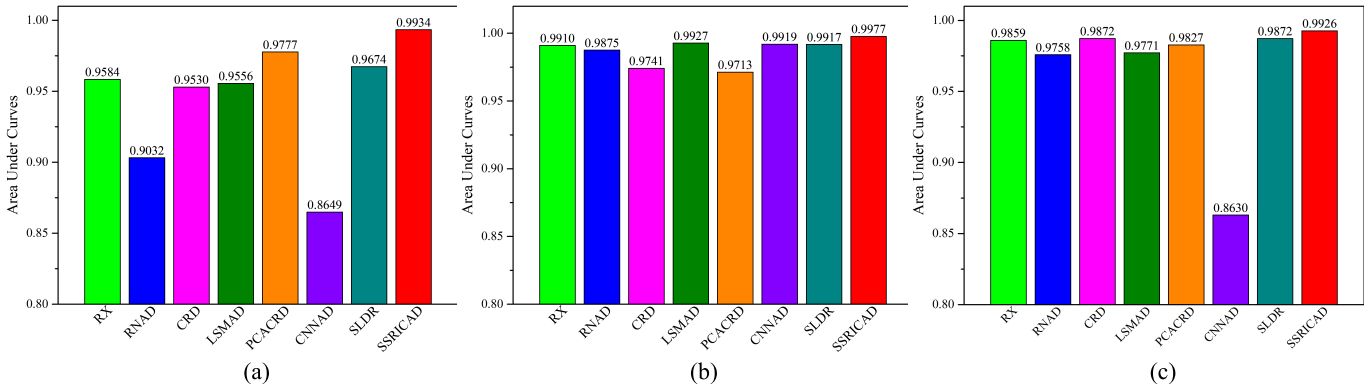


Fig. 11. Comparison of AUC performance for three datasets: (a) Viareggio, (b) Reno Urban, and (c) Avon.

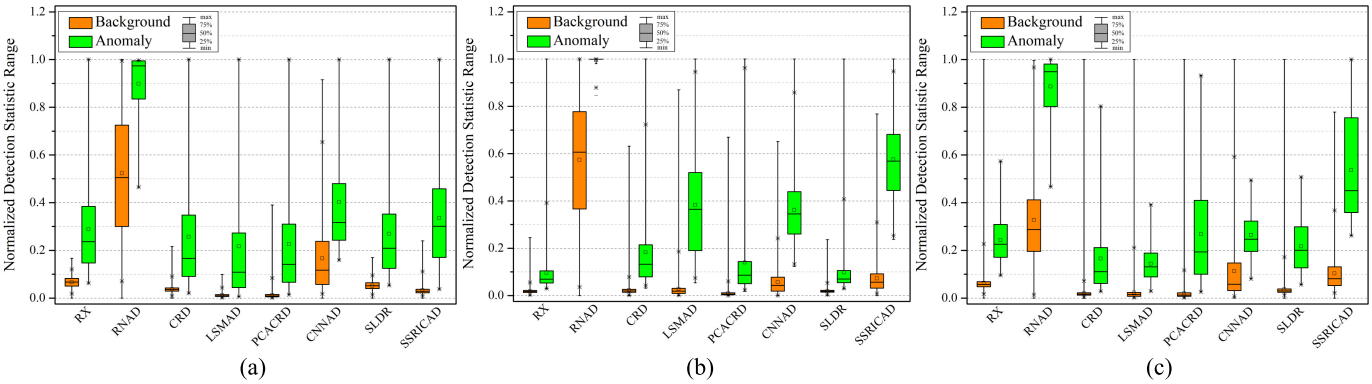


Fig. 12. Comparison of separability performance for three datasets: (a) Viareggio, (b) Reno Urban, and (c) Avon.

TABLE I
AUC VALUES OF ABLATION STUDY FOR THREE DATASETS

Datasets	Viareggio	Reno Urban	Avon
Base	0.9868	0.9607	0.9870
Base+IBCC	0.9911	0.9925	0.9912
Base+ORC	0.9890	0.9924	0.9898
Base+SIC	0.9893	0.9791	0.9879
Base+IBCC+ORC	0.9918	0.9966	0.9914
Base+IBCC+SIC	0.9917	0.9952	0.9911
Base+ORC+SIC	0.9909	0.9956	0.9887
SSRICAD	**0.9934**	**0.9977**	**0.9926**

performance of SSRICAD in background suppression is not that good compared to the four methods, the gap between the two boxes of SSRICAD is much more evident than that of them. For the Avon dataset, RNAD has a large gap between the two columns but fails to suppress the background effectively. The others open small gaps, while SSRICAD opens the gap further, indicating the good separability ability of SSRICAD.

E. Analysis of the Number of Subsets

The parameter analysis of the number of subsets k is discussed in this section. In SSRICAD, we obtain sub-HSIs by a band cross-grouping strategy, the number of subsets determines the number of bands contained in each sub-HSI, which further has an impact on the inherent spectral similarity and redundancy between sub-HSIs. Consequently, we set the number of subsets from 2 to 8 to test AUC performances and

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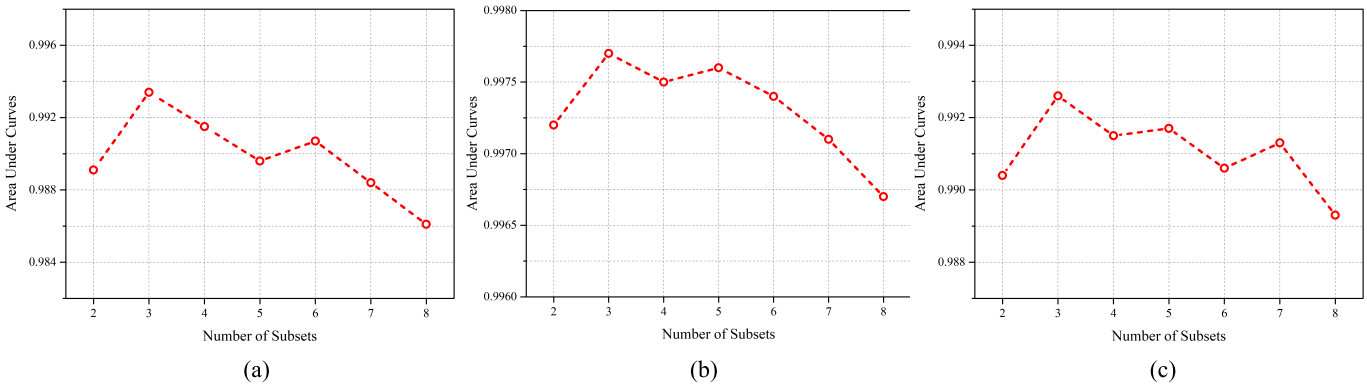


Fig. 13. AUC performance for three datasets with a different number of subsets: (a) Viareggio, (b) Reno Urban, and (c) Avon.

TABLE II
RUNNING TIMES (IN SECONDS) OF DIFFERENT DETECTORS FOR THREE DATASETS

Methods	RX	RNAD	CRD	LSMAD	PCACRD	CNNAD	SLDR	SSRICAD
Viareggio	**0.3616**	10.5656	28.5958	54.4706	13.1881	31.3428	11.7229	15.7839
	±0.045	±0.834	±0.438	±1.248	±0.066	±0.500	±0.888	±0.875
Reno Urban	**0.2772**	14.6211	46.9850	42.6575	27.7033	37.0192	11.7973	14.6771
	±0.021	±0.761	±0.602	±0.583	±0.187	±0.340	±0.916	±0.883
Avon	**0.1949**	17.7216	6.7363	30.4749	3.6618	25.2816	9.5475	13.2263
	±0.036	± 0.693	± 0.101	± 0.318	± 0.059	± 0.398	± 0.719	± 0.764

compare the detection performances. AUC performances with the different number of subsets for the three datasets are shown in Fig. 13.

In general, the AUC values have a trend of rising first and then decreasing. The results indicate that a small number of subsets hardly work on reducing redundancy, while a large number of subsets may destroy spectral similarity and discriminative information for sub-HSIs, which affects the detection results. In this article, we set the number of subsets as 3 for all datasets to achieve excellent detection performances.

F. Ablation Studies of SSRICAD

Ablation experiments are conducted to test the effectiveness of three constraint components of SSRICAD on three datasets in this section. We compare the detection performances under the following eight models: 1) Base: baseNet without constraint; 2) Base + IBCC: baseNet with the IBCC; 3) Base + ORC: baseNet with the ORC; 4) Base + SIC: baseNet with the SIC; 5) Base + IBCC + ORC: baseNet with the IBCC and the ORC; 6) Base + IBCC + SIC: baseNet with the IBCC and the SIC; 7) Base + ORC + SIC: baseNet with the ORC and the SIC; and 8) SSRICAD: baseNet with all three constraints. The numerical results are shown in Table I.

Overall, the AUC values of the models with constraints are higher than those of the Base, which verifies the effectiveness of the three constraint components. The AUC values of SSRICAD are the highest in all three datasets, indicating the necessity of the interband correlation, outlier removal, and SIC, which contribute to improving the detection performance.

G. Analysis of Time Cost

Finally, the running time analysis of different methods for three datasets is provided, as shown in Table II. For a fair comparison, all the methods are implemented under the same condition, focusing on detecting stages. From Table II, the running times of RX are very short for all three datasets, while the representation-based methods, i.e., CRD and PCACRD, are more time-consuming since they need to double calculate the pixels around the central one. Generally, SSRICAD spends more time on detecting anomalies compared to some traditional detectors, but a detection time of around 15 s is acceptable.

IV. CONCLUSION

In this study, we present an SSRICAD method for hyperspectral AD. SSRICAD is formed based on spatial–spectral joint reconstruction and distinguishes anomalies by calculating residuals between the reconstructed image and the Ori-HSI. By imposing an IBCC, SSRICAD utilizes the inherent spectral similarity to improve detection performance while reducing redundancy. To alleviate anomaly contaminations, an ORC is added into networks of HSI reconstruction. Besides, it takes advantage of spatial information extracted from pixel neighbor regions, thus contributing to a more robust detection result. According to the experimental results, the proposed SSRICAD outperforms state-of-the-art methods, presenting a remarkable advantage in identifying anomalies and suppressing background, that is, SSRICAD achieves excellent detection performance with superior separability ability in AD.

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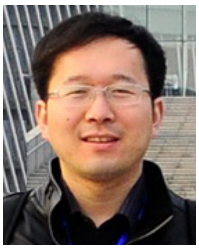
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